

A Minimal Manuscript for Exercising the Paper Review Contract

Paper Review Bench Maintainers

Abstract

This short synthetic manuscript is a complete, self-contained input for an end-to-end paper-review smoke test. It states one elementary claim, gives its derivation, and reports a small consistency check. The scientific content is deliberately modest: the task is to review the manuscript as submitted, not to infer an unstated result or search for outside literature.

1 Claim

For every real number x , the quantity $(x - 1)^2$ is non-negative. Equivalently,

$$x^2 + 1 \geq 2x. \tag{1}$$

We use this identity as a small example of a claim whose statement, argument, and stated limitations can all be inspected directly in the supplied source.

2 Derivation

Expanding the square gives

$$(x - 1)^2 = x^2 - 2x + 1.$$

A square of a real number is non-negative, so rearranging the displayed identity proves Equation 1. No additional assumptions are used.

3 Consistency Check

Table 1 evaluates both sides for three representative inputs. The values are consistent with the claimed inequality.

x	$x^2 + 1$	$2x$
0	1	0
1	2	2
2	5	4

Table 1: A direct finite check of Equation 1.

4 Scope

This manuscript establishes only the elementary inequality in Equation 1. It makes no novelty claim, empirical generalization, or claim about a production system. A useful review can therefore assess whether the claim is stated precisely, the derivation is valid, the check is appropriately interpreted, and the scope is communicated clearly.